

Update Joint Modeling 26/09

Overview Methods:

The three (+1) suggested methods are implemented

1. Classical ML
2. MI + closed form
 1. imputations were done within clusters, 3 imputations per clusters, with a maximum of 10 repetitions.
3. Closed form + iterative reweighting
 1. 30 iterations max
4. LSPIM
 1. **With 100 or more clusters it crashes the system.**, so these are not included
 2. Compared where applicable

Simulation setup

Formula:

$$y_{ij} = \beta_0 + \beta_1 Treatment_i + \beta_2 Time_j + \beta_3 Treatment_i * Time_j + b_{0i} + b_{1i} * Time_j$$

Parameters were taken from the toenail data:

Effect	Parameter	Size
Intercept	β_0	2.4562
Treatment Effect	β_1	0 (<i>0.593 in toenail data (this was a dose effect)</i>)
Time Effect	β_2	0.2792
Treatment x Time	β_3	0.0350 (<i>0 for type I error control</i>)
Random intercept variance	d_{11}	7.3174
Random slope variance	d_{22}	0.2239
Random effects covariance	d_{12}	-0.4985
Residual variance	σ^2	3.1508

Variable	Value
number of clusters	100, 50, 20, 10 (<i>10.000 clusters was asked too, but currently excluded due to large computing time</i>)
number of repeated measures	12
missingness pattern	“half-missing” (half of the clusters have no missingness, other half have uniform probability to have 1-11 observations, dropout is monotone).
number of simulations / scenario	“three observations minimum” (same as half-missing but with three observations minimum for better comparison reweighting) 5.000

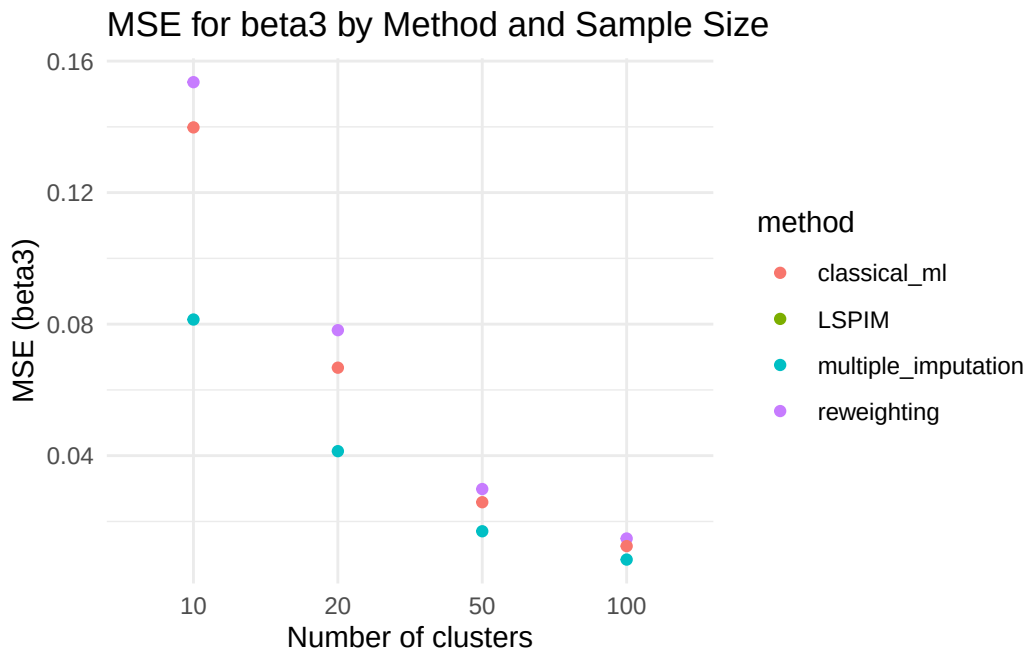
Next steps

1. Asymptotic behavior at 10.000 clusters
2. Improve LSPIM for larger sample sizes?
3. Doublecheck power?

Results - “Half-Missing”

Mean Squared Error

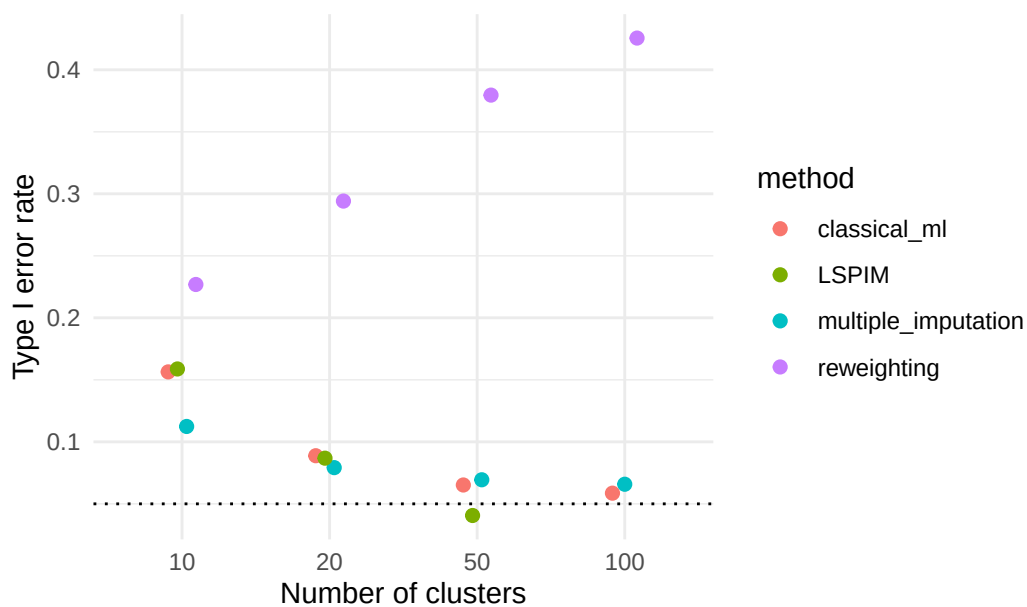
	method	n	mse_beta0	mse_beta1	mse_beta2	mse_beta3
14	classical_ml	100	0.1775848	0.3467385	0.006267283	0.012506696
15	multiple_imputation	100	0.1752215	0.3366275	0.005414892	0.008403148
13	reweighting	100	0.1950869	0.3867126	0.007228470	0.014814955
10	classical_ml	50	0.3385891	0.6786644	0.012934586	0.025844777
9	LSPIM	50	NA	NA	NA	NA
11	multiple_imputation	50	0.3366206	0.6571929	0.011178681	0.017015255
12	reweighting	50	0.3811140	0.7591350	0.015109947	0.029853923
5	classical_ml	20	0.8460474	1.7018867	0.032718379	0.066759908
6	LSPIM	20	NA	NA	NA	NA
7	multiple_imputation	20	0.8425629	1.6580854	0.027085104	0.041375778
8	reweighting	20	0.9523456	1.9236094	0.037892544	0.078166210
2	classical_ml	10	1.7316819	3.4809091	0.069483206	0.139832313
1	LSPIM	10	NA	NA	NA	NA
3	multiple_imputation	10	1.7104991	3.3852958	0.053340899	0.081408696
4	reweighting	10	1.9189127	3.8429596	0.076283988	0.153587543



Type I error control

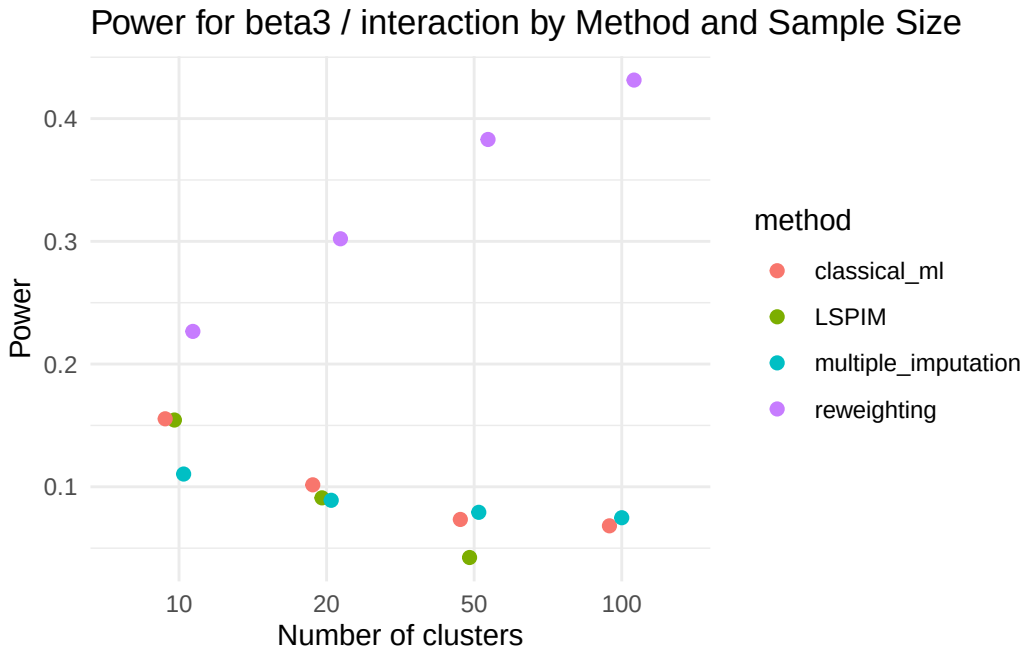
	method	n	alpha
15	classical_ml	100	0.05860000
14	multiple_imputation	100	0.06580000
13	reweighting	100	0.42560000
10	classical_ml	50	0.06520000
11	LSPIM	50	0.04060000
12	multiple_imputation	50	0.06940000
9	reweighting	50	0.37960000
7	classical_ml	20	0.08880000
8	LSPIM	20	0.08681736
5	multiple_imputation	20	0.07920000
6	reweighting	20	0.29415298
1	classical_ml	10	0.15640000
2	LSPIM	10	0.15875876
3	multiple_imputation	10	0.11240000
4	reweighting	10	0.22688564

Type I error rate for beta3 / interaction by Method and Sample :



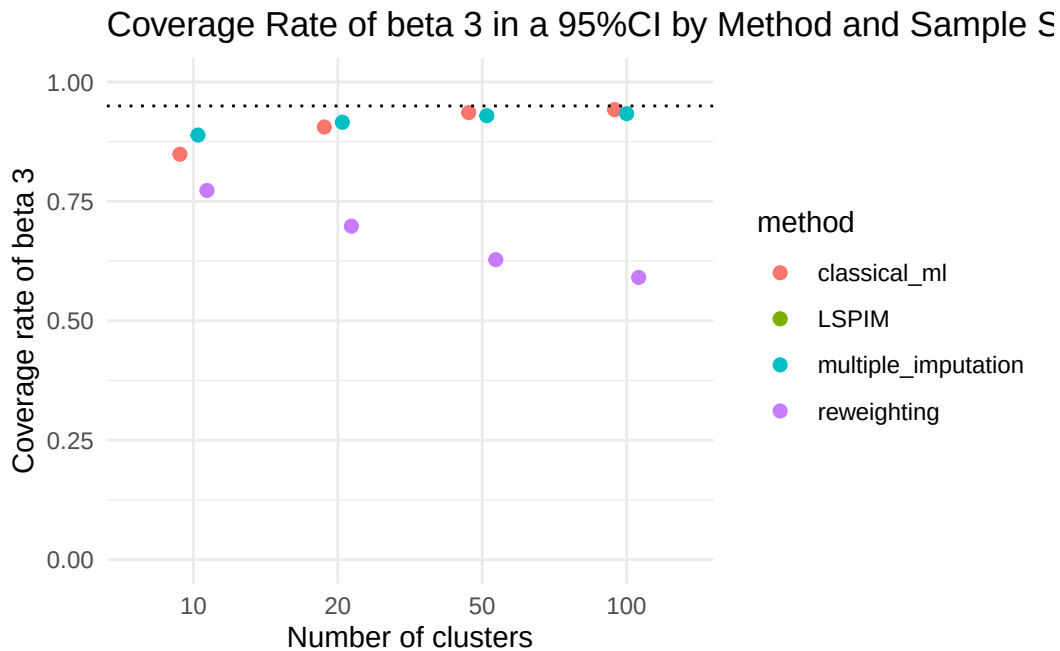
Power

	method	n	power
14	classical_ml	100	0.0682000
15	multiple_imputation	100	0.0748000
13	reweighting	100	0.4314000
10	classical_ml	50	0.0734000
9	LSPIM	50	0.0424000
11	multiple_imputation	50	0.0792000
12	reweighting	50	0.3830000
5	classical_ml	20	0.1016000
6	LSPIM	20	0.0910000
7	multiple_imputation	20	0.0890000
8	reweighting	20	0.3020833
2	classical_ml	10	0.1554000
1	LSPIM	10	0.1544014
3	multiple_imputation	10	0.1104000
4	reweighting	10	0.2266180



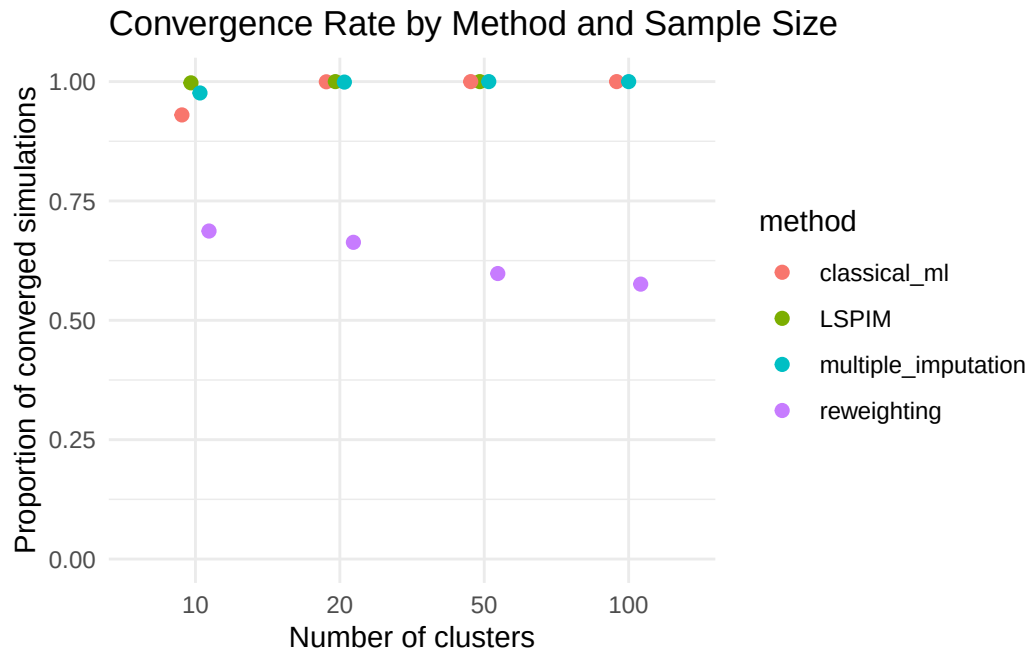
Coverage rate of β_3

	method	n	coverage95_beta3
14	classical_ml	100	0.9422000
15	multiple_imputation	100	0.9338000
13	reweighting	100	0.5908000
10	classical_ml	50	0.9360000
9	LSPIM	50	NA
11	multiple_imputation	50	0.9296000
12	reweighting	50	0.6280000
5	classical_ml	20	0.9058000
6	LSPIM	20	NA
7	multiple_imputation	20	0.9158000
8	reweighting	20	0.6981170
2	classical_ml	10	0.8488000
1	LSPIM	10	NA
3	multiple_imputation	10	0.8890000
4	reweighting	10	0.7729763



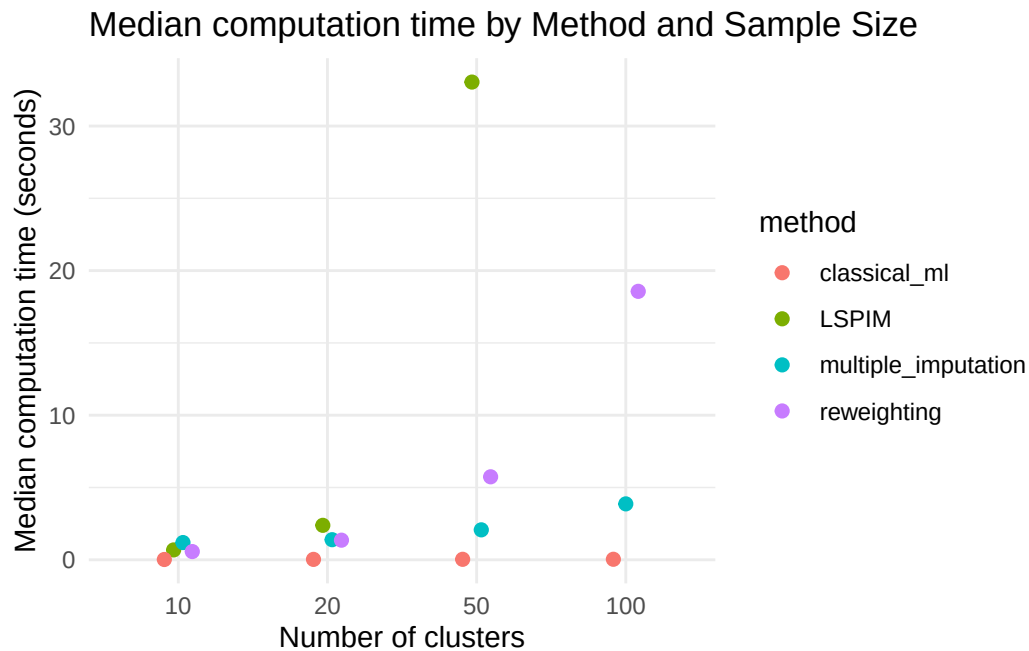
Convergence Rate

	method	n	prop_converged_ok	prop_converged_warning
14	classical_ml	100	0.9494	0.0506
15	multiple_imputation	100	1.0000	0.0000
13	reweighting	100	0.0278	0.5480
10	classical_ml	50	0.9644	0.0356
9	LSPIM	50	0.9770	0.0230
11	multiple_imputation	50	1.0000	0.0000
12	reweighting	50	0.0802	0.5178
5	classical_ml	20	0.9742	0.0254
6	LSPIM	20	0.0000	1.0000
7	multiple_imputation	20	0.9978	0.0012
8	reweighting	20	0.1900	0.4734
2	classical_ml	10	0.8884	0.0418
1	LSPIM	10	0.0002	0.9972
3	multiple_imputation	10	0.9430	0.0332
4	reweighting	10	0.2172	0.4698
	prop_converged_singular			
14		0.0000		
15		0.0000		
13		0.4242		
10		0.0000		
9		0.0000		
11		0.0000		
12		0.4020		
5		0.0004		
6		0.0000		
7		0.0010		
8		0.3366		
2		0.0698		
1		0.0000		
3		0.0238		
4		0.3130		



Computation Time

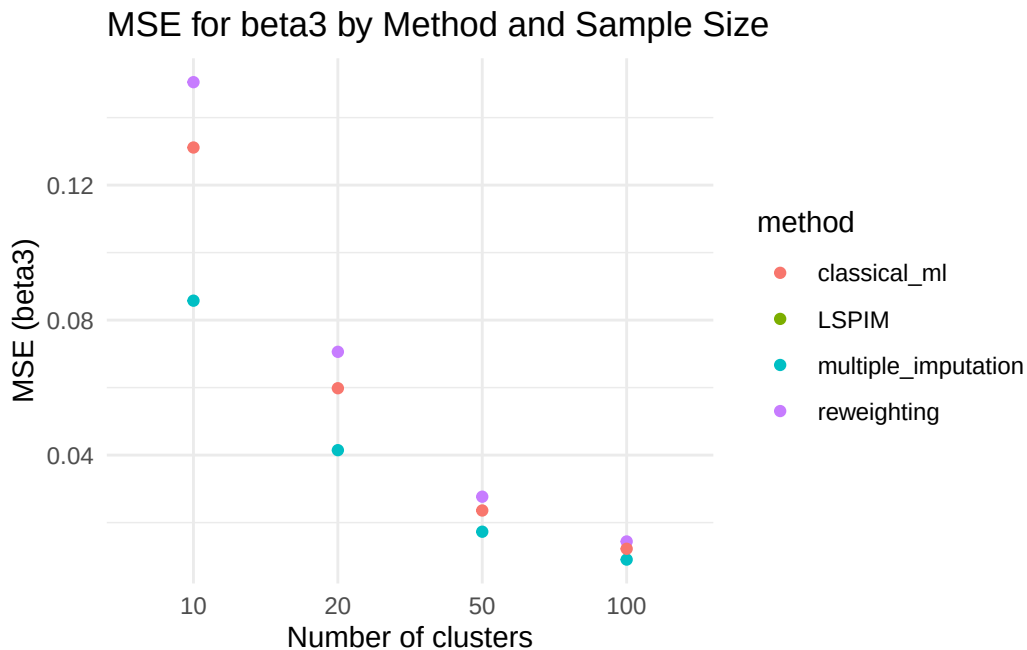
	method	n	time_mean_seconds	time_median_seconds
14	classical_ml	100	0.0410930	0.0360
15	multiple_imputation	100	3.7451656	3.8670
13	reweighting	100	18.0003738	18.5675
10	classical_ml	50	0.0350314	0.0330
9	LSPIM	50	32.9178274	33.0415
11	multiple_imputation	50	2.0250408	2.0720
12	reweighting	50	5.6168340	5.7400
5	classical_ml	20	0.0296850	0.0280
6	LSPIM	20	2.3573492	2.3840
7	multiple_imputation	20	1.3634920	1.3900
8	reweighting	20	1.3253760	1.3570
2	classical_ml	10	0.0223510	0.0210
1	LSPIM	10	0.6730126	0.6820
3	multiple_imputation	10	1.1685812	1.1920
4	reweighting	10	0.5637262	0.5690



Results - “Three observations minimum”

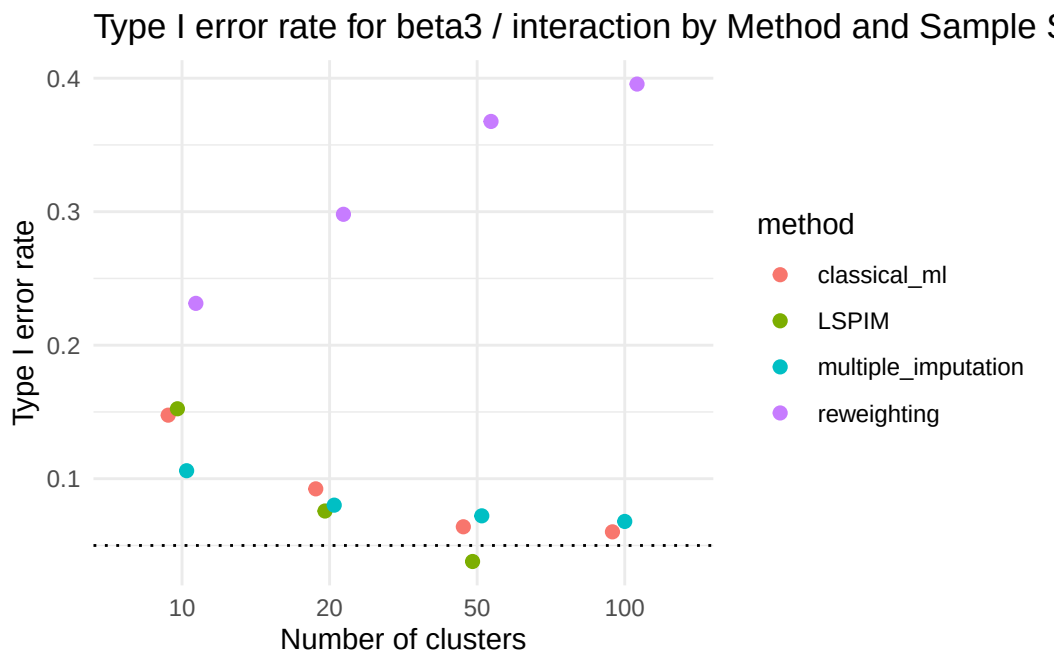
Mean Squared Error

	method	n	mse_beta0	mse_beta1	mse_beta2	mse_beta3
15	classical_ml	100	0.1719896	0.3463735	0.006144953	0.01224748
13	multiple_imputation	100	0.1702289	0.3349660	0.005464958	0.00904226
14	reweighting	100	0.1757522	0.3558946	0.007137799	0.01441028
9	classical_ml	50	0.3414183	0.6941549	0.012011531	0.02359145
10	LSPIM	50	NA	NA	NA	NA
11	multiple_imputation	50	0.3364044	0.6710641	0.010772362	0.01729787
12	reweighting	50	0.3476316	0.7060818	0.013979596	0.02767659
5	classical_ml	20	0.8584329	1.7044704	0.029562476	0.05981631
6	LSPIM	20	NA	NA	NA	NA
7	multiple_imputation	20	0.8435923	1.6502020	0.025519679	0.04146113
8	reweighting	20	0.8758266	1.7388715	0.034487186	0.07060698
4	classical_ml	10	1.7320312	3.4259817	0.064273494	0.13114268
1	LSPIM	10	NA	NA	NA	NA
2	multiple_imputation	10	1.6975273	3.3247218	0.052766464	0.08575475
3	reweighting	10	1.7690309	3.5067844	0.074951793	0.15050155



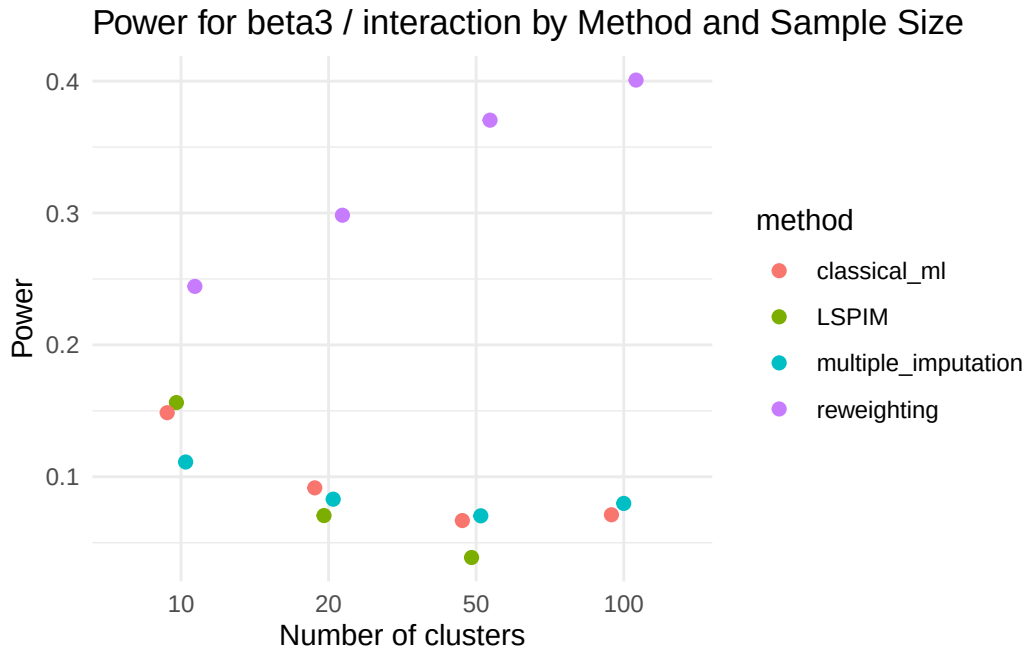
Type I error control

	method	n	alpha
13	classical_ml	100	0.0602000
14	multiple_imputation	100	0.0680000
15	reweighting	100	0.3956000
9	classical_ml	50	0.0640000
10	LSPIM	50	0.0380000
12	multiple_imputation	50	0.0722000
11	reweighting	50	0.3676000
5	classical_ml	20	0.0924000
6	LSPIM	20	0.0758000
7	multiple_imputation	20	0.0802000
8	reweighting	20	0.2980596
1	classical_ml	10	0.1476000
2	LSPIM	10	0.1523829
3	multiple_imputation	10	0.1060000
4	reweighting	10	0.2313433



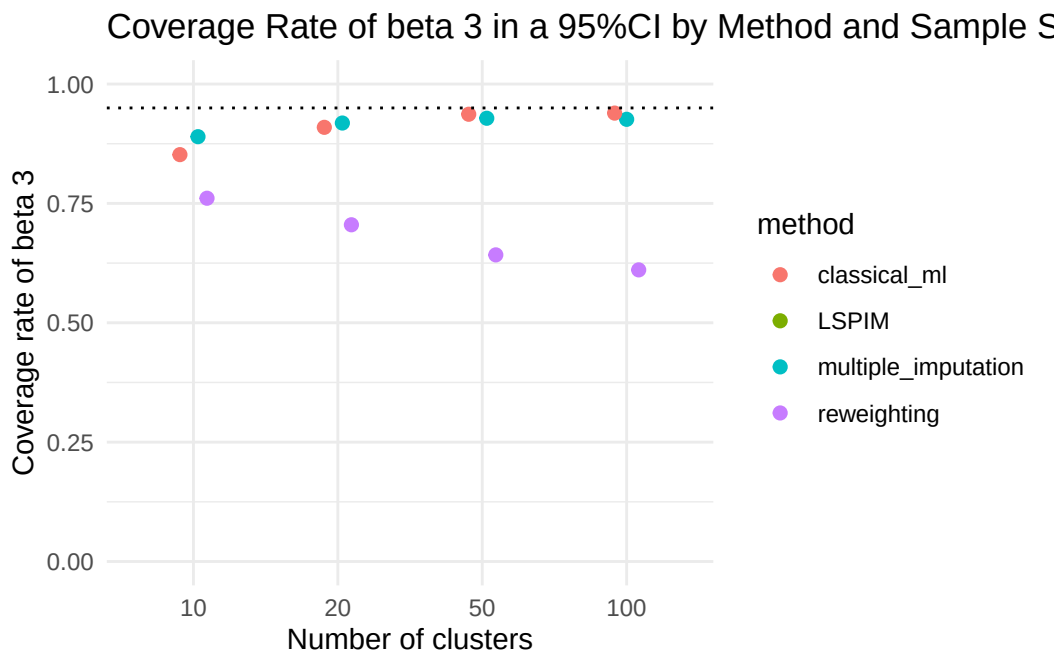
Power

	method	n	power
15	classical_ml	100	0.0712000
13	multiple_imputation	100	0.0798000
14	reweighting	100	0.4008000
9	classical_ml	50	0.0668000
10	LSPIM	50	0.0388000
11	multiple_imputation	50	0.0704000
12	reweighting	50	0.3704000
5	classical_ml	20	0.0916000
6	LSPIM	20	0.0706000
7	multiple_imputation	20	0.0830000
8	reweighting	20	0.2983193
4	classical_ml	10	0.1486000
1	LSPIM	10	0.1562625
2	multiple_imputation	10	0.1112000
3	reweighting	10	0.2443594



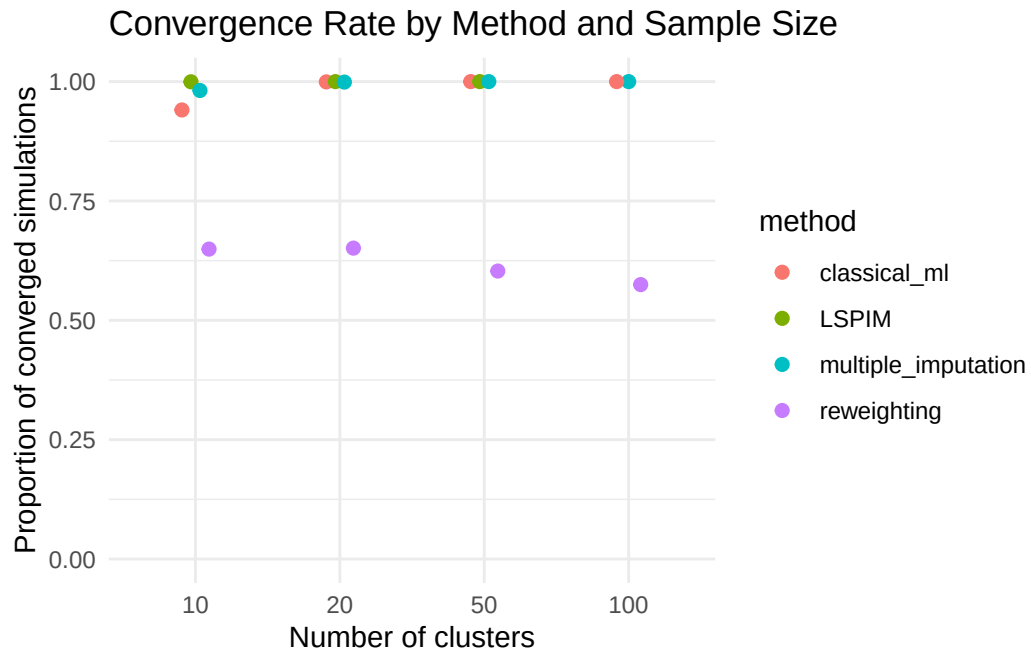
Coverage rate of β_3

	method	n	coverage95_beta3
15	classical_ml	100	0.9392000
13	multiple_imputation	100	0.9262000
14	reweighting	100	0.6108000
9	classical_ml	50	0.9368000
10	LSPIM	50	NA
11	multiple_imputation	50	0.9284000
12	reweighting	50	0.6422000
5	classical_ml	20	0.9094000
6	LSPIM	20	NA
7	multiple_imputation	20	0.9184000
8	reweighting	20	0.7052821
4	classical_ml	10	0.8522000
1	LSPIM	10	NA
2	multiple_imputation	10	0.8898000
3	reweighting	10	0.7608783



Convergence Rate

	method	n	prop_converged_ok	prop_converged_warning
15	classical_ml	100	0.9452	0.0548
13	multiple_imputation	100	1.0000	0.0000
14	reweighting	100	0.0068	0.5680
9	classical_ml	50	0.9722	0.0278
10	LSPIM	50	0.9918	0.0082
11	multiple_imputation	50	1.0000	0.0000
12	reweighting	50	0.0434	0.5598
5	classical_ml	20	0.9794	0.0200
6	LSPIM	20	0.0002	0.9998
7	multiple_imputation	20	0.9982	0.0008
8	reweighting	20	0.1312	0.5200
4	classical_ml	10	0.9034	0.0372
1	LSPIM	10	0.0000	0.9996
2	multiple_imputation	10	0.9584	0.0230
3	reweighting	10	0.1792	0.4700
	prop_converged_singular			
15			0.0000	
13			0.0000	
14			0.4252	
9			0.0000	
10			0.0000	
11			0.0000	
12			0.3968	
5			0.0006	
6			0.0000	
7			0.0010	
8			0.3488	
4			0.0594	
1			0.0000	
2			0.0186	
3			0.3508	



Computation Time

	method	n	time_mean_seconds	time_median_seconds
15	classical_ml	100	0.0434544	0.0350
13	multiple_imputation	100	3.6934174	3.8170
14	reweighting	100	20.9403360	21.4160
9	classical_ml	50	0.0361734	0.0330
10	LSPIM	50	38.8267218	38.8575
11	multiple_imputation	50	1.9969216	2.0590
12	reweighting	50	6.2693764	6.4605
5	classical_ml	20	0.0279068	0.0270
6	LSPIM	20	2.5706270	2.6060
7	multiple_imputation	20	1.3302164	1.3590
8	reweighting	20	1.4594732	1.4850
4	classical_ml	10	0.0229780	0.0210
1	LSPIM	10	0.7117686	0.7200
2	multiple_imputation	10	1.1549174	1.1750
3	reweighting	10	0.6087186	0.6170

